



QUANTUM BLUFF

Technical Design Report

Evolving Texas Hold'em - Desktop to Online

Project Manager: Omar
Requirements Deadline: February 13, 2025
Total Duration: 16 weeks

1 Technical Choices

1.1 Languages and Technologies

Component	Technology	Rationale
Backend	Java 17 + Spring Boot	Performance, stability, multithreading
Desktop Client	C# WPF	Rich interface, Windows integration
Database	PostgreSQL 15	ACID transactions, reliability
Communication	WebSocket + JSON	Real-time, standardized
Security	TLS 1.3 + JWT	Strong encryption, authentication

2 New Rules - Hidden Bets

2.1 Core Mechanics

- **Phase:** Bets only pre-flop
- **Revelation:** At showdown only

- **Bet Types:**
 - Hand winner
 - Specific formation (Full House, Flush, etc.)
 - High card combinations
- **Odds:** Calculated based on predefined probabilities

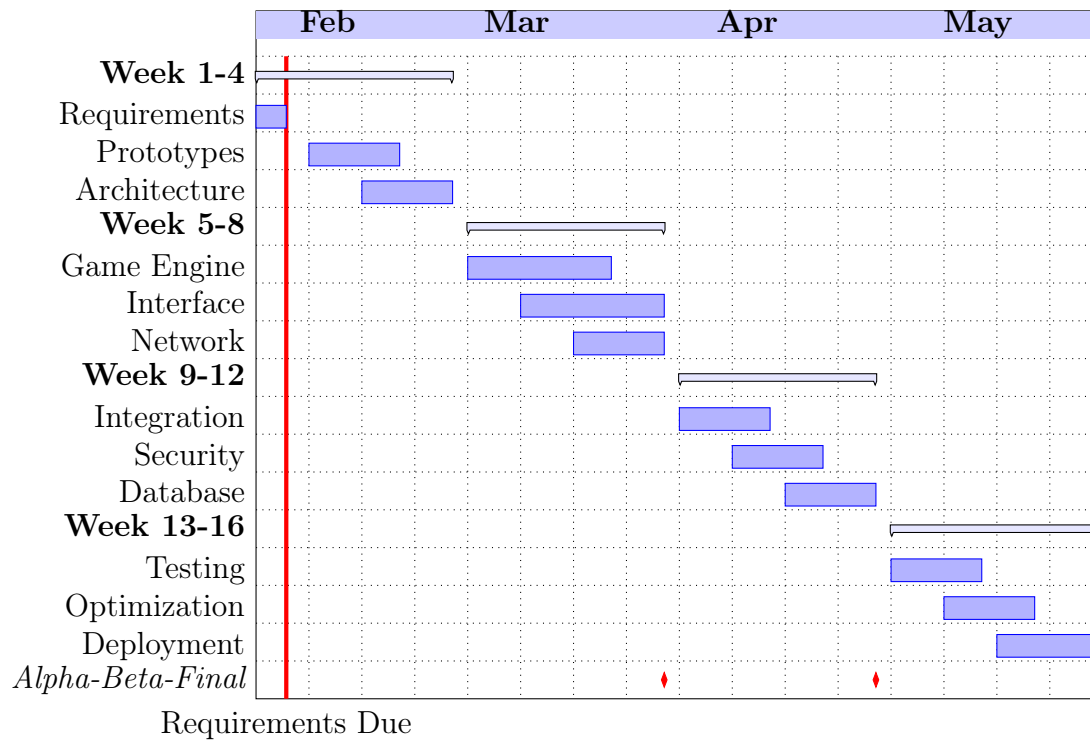
2.2 Bet Examples

Bet Type	Condition	Odds
Full House	Make a Full House	1:8
Flush	Make a Flush	1:5
High Cards	A-K in hand	1:3

3 Team Distribution

tablehead Team	Members	Main Missions
Game Engine	Azra, Souheil	• Texas Hold'em logic • Hidden Bets algorithms • Probability calculations
Interface	Mohamed, Yigit	• Desktop GUI • Card/chip animations • Simulated payment gateway
Network	Massi, Linda	• Socket communication • Multi-room management • IP geolocation
Data	Abdoulae	• SQL database schema • Daily Rewards system • Player statistics
Management	Omar	• Project management • Continuous integration • Quality assurance

4 Gantt Chart - 16 Weeks



5 Key Points by Phase

5.1 Week 1-4: Design

- Requirements document delivery (Feb 13)
- Hidden Bets rules validation
- Final technical choices
- System architecture

5.2 Week 5-8: Core Development

- Functional game engine
- Basic desktop interface
- Client-server communication

5.3 Week 9-12: Integration

- Alpha release (Apr 4)
- Hidden Bets integrated
- Security and authentication

5.4 Week 13-16: Finalization

- Beta release (May 2)
- Intensive testing
- Performance optimization
- Final deployment (May 31)

6 Identified Risks

Risk	Mitigation
Hidden Bets complexity	Rapid prototyping, unit testing
Network performance	Packet optimization, compression
Payment security	TLS encryption, server validation
Deadline respect	Daily standups, Gantt tracking

7 Conclusion

The Quantum Bluff project is structured over 16 weeks with clear milestones. The main innovation lies in Hidden Bets, a unique mechanic adding depth to traditional Texas Hold'em. The team is distributed according to skills with well-defined responsibilities.

Critical Milestones:

- February 13: Requirements document
- April 4: Alpha version
- May 2: Beta version
- May 31: Final version